

Experiment : 4

Statistical Hypothesis Testing Using Python

Mapped CO(s) : CO2

Aim

To perform statistical hypothesis testing on the Titanic dataset using Python and interpret the results using p-values.

Objectives

- To understand the concept of hypothesis testing.
- To formulate null and alternative hypotheses.
- To perform statistical tests using Python.
- To interpret p-values and significance levels.
- To draw conclusions based on the test results.

Prerequisites

- Basic knowledge of Python programming.
- Basic understanding of probability and statistics.
- Python 3.x installed.
- Jupyter Notebook or Google Colab.
- Required libraries: Pandas, NumPy, SciPy, Matplotlib, and Seaborn.

Theory/Background

Statistical Hypothesis Testing is a method used to determine whether there is sufficient evidence to support a claim about a population based on sample data.

A hypothesis test consists of two hypotheses:

- **Null Hypothesis (H_0):** Assumes there is no significant difference or relationship between variables.
- **Alternative Hypothesis (H_1):** Assumes there is a significant difference or relationship between variables.

The **p-value** is compared with the significance level ($\alpha = 0.05$).

- If **p-value < 0.05**, reject the null hypothesis.
- If **p-value $\geq$ 0.05**, fail to reject the null hypothesis.

In this experiment, an **Independent Sample t-test** is performed to determine whether the average age of passengers differs significantly based on survival status.

Algorithm/Methodology

- Import the required Python libraries.
- Load the Titanic dataset.
- Select the required variables.
- Remove missing values.
- Separate passengers into survived and not survived groups.
- Perform an Independent Sample t-test.
- Display the t-statistic and p-value.
- Interpret the hypothesis test result.

Dataset Description

Dataset Name: Titanic Dataset

Source: Seaborn / Kaggle

The Titanic dataset contains passenger information such as age, gender, passenger class, fare, and survival status. This experiment uses the **Age** and **Survived** variables to perform hypothesis testing.

Attributes Used

Attribute	Description
survived	Survival status (0 = No, 1 = Yes)
age	Age of passenger

Program/Implementation

The program should include:

- Importing required libraries.
- Loading the Titanic dataset.
- Cleaning missing values.
- Separating the two groups.
- Performing the Independent Sample t-test using SciPy.
- Displaying the t-statistic and p-value.
- Interpreting the result.

Test Cases

Test Case	Expected Result
Load Titanic dataset	Dataset loads successfully
Remove missing values	Dataset cleaned successfully
Perform t-test	t-statistic and p-value displayed
Interpret result	Accept or reject the null hypothesis

Experimental Results

Observation

- The Titanic dataset was successfully loaded.
- Missing values in the Age column were removed.
- Passengers were divided into survived and non-survived groups.
- An Independent Sample t-test was successfully performed.
- The p-value indicated whether the difference in average age between the two groups was statistically significant.

Result

Statistical hypothesis testing was successfully performed using Python. The hypothesis was tested using an Independent Sample t-test, and the conclusion was drawn based on the calculated p-value.

Conclusion

The experiment demonstrated the application of statistical hypothesis testing using Python. The Independent Sample t-test was used to compare the mean age of survived and non-survived passengers. Based on the p-value, an appropriate statistical conclusion was made regarding the null hypothesis. Hypothesis testing is an important tool for making data-driven decisions and validating assumptions in data science.

Exercises/Assignments

1. Perform a t-test using the **Fare** variable instead of **Age**.
2. Change the significance level from **0.05** to **0.01** and compare the result.
3. Interpret the obtained p-value.
4. Apply another statistical test such as the Chi-Square test.
5. State the null and alternative hypotheses for a different variable.